ServiceLens

Analysing Customer Satisfaction Drivers and Support Interaction Patterns Using PySpark and Tableau

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Introduction

Customer satisfaction (CSAT) is a critical indicator of service quality and organisational performance. In customer support environments, satisfaction outcomes influence retention, loyalty, operational efficiency, and long-term business success. The widespread adoption of digital support systems has produced large volumes of operational data through help desks, ticketing platforms, CRM systems, and feedback mechanisms.

Traditional service quality research identifies responsiveness as a key determinant of customer satisfaction. However, recent studies suggest that customer effort, issue resolution quality, and communication effectiveness may be equally important drivers. Contemporary services marketing research also emphasises managing the complete customer journey rather than isolated service encounters (Zeithaml et al., 2018). This project proposes **ServiceLens**, a customer analytics framework combining PySpark-based analytics with Tableau visualisation to investigate satisfaction drivers and support interaction patterns through classification, clustering, and regression techniques.

Problem Statement & Research Aim

Customer support teams routinely collect data on ticket management, response times, escalations, and customer feedback, yet struggle to transform these records into actionable insights. A common assumption is that reducing response time automatically improves satisfaction. Emerging research challenges this, suggesting issue complexity, customer effort, and resolution effectiveness are equally significant.

The selected dataset contains ≈85,000 customer support records. Although this does not meet the strict definition of Big Data, it represents a realistic operational dataset that provides an opportunity to demonstrate scalable analytics methodologies using PySpark.

Research Aim: To investigate the drivers of customer satisfaction and discover customer support interaction patterns through scalable analytics using PySpark and Tableau.

Dataset

Source: Kaggle Customer Support Ticket Dataset
Link: `https://www.kaggle.com/datasets/akashbommidi/customer-support-data`
Size: Approximately 85,000 records — structured tabular format with satisfaction indicators, ticket lifecycle information, and operational performance metrics.

Key Variables: Response Time, Resolution Time, Ticket Priority, Ticket Category, Interaction Count, Support Channel, Ticket Status, Customer Satisfaction Score.

Research Questions & Hypotheses

RQ1: Which operational factors have the strongest influence on customer satisfaction?
RQ2: Can customer satisfaction be accurately predicted using support interaction data?
RQ3: What support interaction profiles emerge through unsupervised learning?
RQ4: What is the relative contribution and effect size of response time, resolution efficiency, interaction frequency, escalation events, and customer effort on satisfaction outcomes?

RQ1 identifies influential variables, whereas RQ4 specifically quantifies the magnitude, direction, and practical significance of those influences.

H1: Faster response and resolution times are positively associated with customer satisfaction.
H2: Resolution effectiveness and customer effort exert greater influence than response speed alone.
H0: No statistically significant relationship exists between operational support metrics and satisfaction.

The competing hypotheses allow empirical evaluation of whether responsiveness remains the dominant explanatory factor predicted by traditional service quality models.

Literature Foundation

The conceptual basis is the **SERVQUAL** framework (Parasuraman et al., 1988), which identifies reliability, responsiveness, assurance, empathy, and tangibles as core service quality dimensions. Responsiveness is frequently cited as a major satisfaction determinant.

However, subsequent research in digital service environments suggests customer satisfaction is influenced by a broader set of factors including perceived customer effort, communication effectiveness, and issue resolution quality (Dixon et al., 2010; Lemon & Verhoef, 2016). **Customer Effort Theory** proposes that reducing customer effort often has a greater impact on loyalty than exceeding expectations (Dixon et al., 2010) — directly informing the Customer Effort Score Proxy in Section 7. Managing the complete service journey rather than isolated response metrics is also emphasised (Lemon & Verhoef, 2016; Zeithaml et al., 2018).

Justification for PySpark

PySpark is selected primarily for architectural and methodological reasons rather than computational necessity. It provides distributed data processing, scalable feature engineering, ML pipeline construction, enterprise-oriented design, and future scalability to larger datasets. Although the current dataset contains ≈85,000 records, the project demonstrates how customer support analytics could be operationalised within production-scale environments — functioning as a *proof-of-concept* for scalable enterprise analytics.

Feature Engineering Strategy

- **Response Efficiency Ratio** — Responsiveness relative to ticket complexity. *Rationale:* Customers often interpret rapid engagement as a signal of service quality and attentiveness.
- **Resolution Efficiency Ratio** — Effectiveness of ticket resolution relative to communication effort. *Rationale:* Successful resolution may contribute more strongly to satisfaction than speed alone.
- **Interaction Density** — Communication intensity throughout the ticket lifecycle. *Rationale:* High density may indicate proactive engagement; conversely, excessive communication without resolution may signal inefficiency. Direction evaluated empirically.
- **Escalation Indicator** — Binary variable for whether ticket escalation occurred. *Rationale:* Escalations often reflect issue complexity, operational challenges, or customer dissatisfaction.
- **Customer Effort Score Proxy** — Derived from interaction frequency and resolution duration. *Rationale:* Grounded in Customer Effort Theory (Dixon et al., 2010); enables empirical testing of whether effort-related factors outperform responsiveness metrics in predicting satisfaction.

Methodology

Phase 1 — Data Preparation

Data cleaning, missing value treatment, outlier detection, data transformation, and feature engineering. *Tools: PySpark, Pandas.*

Phase 2 — Exploratory Data Analysis

Distribution analysis, correlation analysis, satisfaction trend exploration, and operational performance exploration. *Tools: PySpark, Tableau.*

Phase 3 — Classification

Goal: Predict customer satisfaction outcomes.
Algorithms: Logistic Regression, Random Forest, Gradient Boosted Trees (PySpark MLlib).
Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC.

Phase 4 — Clustering

Goal: Discover customer support interaction profiles.
Algorithms: K-Means, Bisecting K-Means (PySpark MLlib).
Hypothesised Profiles: Efficient Resolution Customers; Communication Intensive Customers; Dissatisfied High-Effort Customers. These are exploratory hypotheses validated empirically through clustering results.
Metrics: Silhouette Score, Davies-Bouldin Index.

Phase 5 — Regression Analysis

Goal: Estimate variable influence and effect size.
Algorithms: Linear Regression, Random Forest Regression.
Outputs: Variable importance rankings, effect size estimation, sensitivity analysis, partial dependence interpretation.

Ethical Considerations

The study follows responsible data analytics principles: removal of personally identifiable information; fairness assessment across demographic groups where data permits; transparent reporting of model performance; responsible interpretation of behavioural patterns; and avoidance of discriminatory conclusions.

Limitations

- Limited generalisability beyond the selected dataset and industry context.
- Proxy-based measurement of customer effort rather than direct perceptual data.
- Observational data cannot establish definitive causal relationships.
- Dataset size does not represent true large-scale distributed Big Data systems.

Expected Outcomes

The project is expected to produce a ranked assessment of satisfaction drivers, predictive satisfaction models, customer support interaction profiles, interactive Tableau dashboards, and evidence-based service improvement recommendations.

Importantly, a null or contrary finding would also provide meaningful insight. If responsiveness is not identified as the dominant satisfaction driver, results would challenge conventional assumptions and support alternative strategies centred on effort reduction and resolution quality.

Project Plan

Week	Activity
1	Literature Review and Research Design
2	Data Understanding and Cleaning
3	Feature Engineering
4	Exploratory Data Analysis
5	Classification Modelling
6	Model Evaluation and Refinement
7	Clustering Analysis
8	Regression Analysis
9	Tableau Dashboard Development
10	Report Writing and Final Validation

Contingency Plan: If model performance is unsatisfactory, additional feature engineering will be conducted, alternative algorithms evaluated, hyperparameter optimisation performed, and feature selection methods applied. Findings will be reported transparently regardless of predictive performance.

Conclusion

ServiceLens proposes a scalable analytics framework for investigating customer satisfaction drivers and support interaction patterns. By integrating PySpark, machine learning, and Tableau visualisation, the project aims to generate actionable insights while critically evaluating whether responsiveness remains the dominant determinant of customer satisfaction in contemporary digital support environments.

References

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